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Education for Future Generations

AI Course

Team Project Final Report

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Joan Sanchis Sais
Jorge Andres Lopez Vazquez

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Abstract

Recent advances in the use of electroencephalogram (EEG) data for the diagnosis of neurodegenerative diseases have opened new avenues for early detection and classification of dementia. In this project, we present a neural network-based system and a user-friendly application capable of analyzing resting-state EEG signals to predict both the Mini-Mental State Examination (MMSE) score and the probable diagnostic group—Alzheimer's disease, frontotemporal dementia, or healthy control. The model was trained using a high-quality, publicly available EEG dataset comprising 88 subjects (36 with Alzheimer's disease, 23 with frontotemporal dementia, and 29 healthy controls), recorded with 19-channel scalp electrodes in a resting-state condition with eyes closed. Signal preprocessing included artifact subspace reconstruction and independent component analysis to ensure denoising and signal reliability. The resulting tool offers an accessible, non-invasive solution for clinicians and researchers, providing fast and interpretable predictions of cognitive impairment severity. This system contributes to the growing field of AI-assisted neurology by supporting early diagnosis, patient monitoring, and deeper understanding of neurodegenerative processes through EEG analysis.

Introduction

Objective

Our goal is to assist healthcare professionals and researchers in the early detection and assessment of neurodegenerative diseases by providing an user-friendly and AI-powered tool that enables us to predict a reliable grade of cognitive impairment using resting-state EEG data.

Motivation

Neurodegenerative diseases such as Alzheimer's and frontotemporal dementia affect millions of people worldwide and are increasing in prevalence due to an aging population. Early and accurate diagnosis is essential for treatment planning, clinical trials, and improving quality of life, yet current diagnostic methods are often expensive, invasive, or subjective. EEG is a non-invasive, low-cost, and widely available technology, but its analysis can be complex and time-consuming. By leveraging machine learning and deep neural networks, we aim to simplify and

automate EEG interpretation to support medical decision-making and research efforts.

Product Summary

What have you done? How do you use it? How does it relate to your objective? What are the functionalities and benefits enabled by AI?

We have developed a neural network model and a web application capable of analyzing resting-state EEG recordings to predict a subject's Mini-Mental State Examination (MMSE) score and classify the subject into one of three diagnostic groups: Alzheimer's disease, frontotemporal dementia, or healthy control. The tool uses EEG data formatted in the BIDS standard. Preprocessing includes advanced denoising techniques such as Artifact Subspace Reconstruction and Independent Component Analysis. The application provides fast, interpretable outputs through an intuitive user interface, making it accessible to non-expert users. The integration of AI enables automated cognitive health assessments, facilitates longitudinal monitoring, and supports large-scale clinical studies.

Background

The electroencephalogram (EEG) is a widely used neurophysiological tool that offers real-time insights into brain activity. In recent years, there has been growing interest in applying machine learning (ML) and deep learning techniques to EEG data for the early diagnosis and monitoring of neurodegenerative diseases. Resting-state EEG, in particular, provides a non-invasive, low-cost, and accessible method to detect abnormal brain patterns linked to conditions like Alzheimer's and frontotemporal dementia. However, progress is slowed by the limited availability of public EEG datasets labeled with clinical scores such as the Mini-Mental State Examination (MMSE), and by the complexity of manual EEG analysis, which requires expert knowledge and is not scalable.

The scale of this domain is substantial, with over 55 million people worldwide currently affected by dementia—a number expected to triple by 2050. There is an urgent need for scalable, objective diagnostic tools. Our project uses a publicly available dataset of 88 resting-state EEG recordings, including MMSE scores, to

train an AI system that predicts cognitive impairment and classifies diagnostic groups. This approach demonstrates the potential of AI-enhanced EEG analysis to support early diagnosis and improve accessibility to dementia screening, especially in under-resourced settings.

Related Work

Several studies have applied machine learning to resting-state EEG for diagnosing neurodegenerative diseases, but traditionally, EEG analysis has been done manually by experts, which is time-consuming and subjective. Public datasets with clinical labels like MMSE are rare, limiting model development. Our work leverages a publicly available EEG dataset including Alzheimer's and frontotemporal dementia cases, providing both classification and MMSE prediction, along with an easy-to-use application that integrates our neural network for practical clinical use.

Product

Our product is an intuitive application designed to assist healthcare professionals in assessing cognitive health through EEG analysis. Users simply upload a resting-state EEG recording, and the system quickly analyzes the data to provide two key outputs: an estimated Mini-Mental State Examination (MMSE) score indicating the level of cognitive impairment, and a classification of the subject into one of three groups — Alzheimer's disease, frontotemporal dementia, or healthy control. The tool offers clear visualizations and reports, making the results easy to interpret and integrate into clinical workflows. Its user-friendly interface and fast processing time enable routine use in both clinical and research settings, enhancing early diagnosis and monitoring of neurodegenerative diseases.

Market

Our primary target audience consists of neurologists, geriatricians, and clinical neurophysiologists who diagnose and monitor neurodegenerative diseases such as Alzheimer's and frontotemporal dementia. These specialists require reliable, non-invasive tools that can support early diagnosis and track disease progression efficiently.

Secondary users include researchers working on dementia and brain disorders who need standardized, accessible EEG analysis tools to accelerate their studies.

Product Description & Usage

What goes in?

The user uploads a resting-state EEG recording collected with a standard 19-electrode system (in BIDS or common EEG format). Additionally, the user inputs the patient's age and sex through the application interface. The EEG is recorded while the patient is resting with eyes closed.

What comes out?

The system outputs:

1. An estimated Mini-Mental State Examination (MMSE) score, indicating the cognitive impairment level.
2. A classification label categorizing the EEG as Alzheimer's disease, frontotemporal dementia, or healthy control.

Typical user interaction flow:

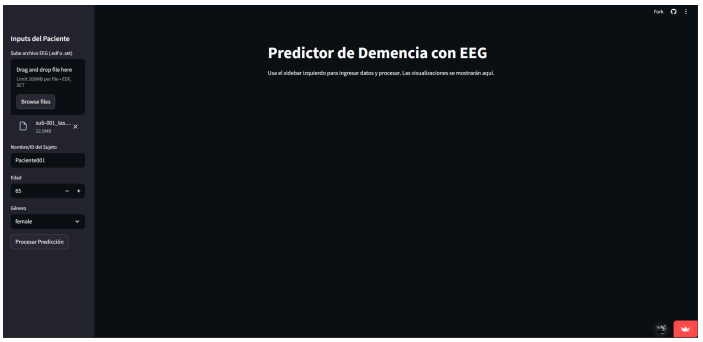
1. The clinician or researcher logs into the app.
2. They upload the raw EEG recording file.
3. They enter the patient's age and sex.
4. The AI neural network processes the EEG data along with age and sex to generate predictions.
5. The predicted MMSE score and diagnosis classification are displayed.

Where AI comes in:

AI is applied in step 4, where the neural network analyzes the combined input of raw EEG signals, age, and sex to predict cognitive decline severity and disease classification with accuracy and speed.

Demo's Prototypes

Final application



Results

Data

We used a publicly available resting-state EEG dataset containing 88 subjects: 36 diagnosed with Alzheimer's disease, 23 with frontotemporal dementia, and 29 healthy controls. The EEG signals were recorded using a 19-electrode monopolar montage during eyes-closed resting state. Each subject's Mini-Mental State Examination (MMSE) score was included as a clinical label. The dataset was obtained from A Dataset of Scalp EEG Recordings of Alzheimer's Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG. An example input consists of a raw EEG recording (approximately 5 minutes) alongside the patient's age, sex, and MMSE score as ground truth.

Modeling

We used a multitask feedforward neural network implemented in TensorFlow/Keras.

The model was designed to perform two tasks simultaneously:

1. Classification into one of three diagnostic groups (Alzheimer's disease, frontotemporal dementia, or healthy control).
2. Regression to predict the patient's Mini-Mental State Examination (MMSE) score.

Why this model?

We chose a multitask approach because classification and regression are related tasks that benefit from shared EEG feature representations. This architecture allows the model to generalize better by learning from both tasks at once, especially when working with a relatively small dataset. It also reduces the need for training two separate models.

Model architecture

- **Input:** a feature vector derived from EEG recordings, concatenated with age and sex (numerically encoded).
- **Shared layers:** 3 fully connected layers with ReLU activations, batch normalization, and dropout for regularization.
- **Classification head:** 2 dense layers followed by a softmax layer with 3 output neurons (one per class).
- **Regression head:** 2 dense layers followed by a linear output neuron (predicting MMSE score).

Framework and training details

- Framework: TensorFlow 2.0 using the Keras Functional API.
- Optimizer: Adam, with default learning rate.

- Loss functions:
    - Categorical Crossentropy for the classification output.
    - Mean Squared Error (MSE) for the regression output.

- We trained the model for up to 100 epochs using early stopping (patience=10) and ReduceLROnPlateau to prevent overfitting.
- The data was split into training, validation, and test sets using stratified sampling on the class labels to preserve class balance.

With this setup, the model learns to predict both cognitive state and clinical grouping from the same EEG-derived input, making it efficient and clinically relevant.

Evaluation

Our multi-task neural network performs well on both classification and regression tasks. The model was trained to simultaneously classify patients into one of three diagnostic groups (Alzheimer's disease, frontotemporal dementia, and healthy controls) and predict their MMSE cognitive score.

Classification Performance:

- Validation accuracy reached approximately 96%, with a tight 95% bootstrap confidence interval around this value, indicating strong model generalization.
- The classification loss consistently decreased during training, and no overfitting was observed.

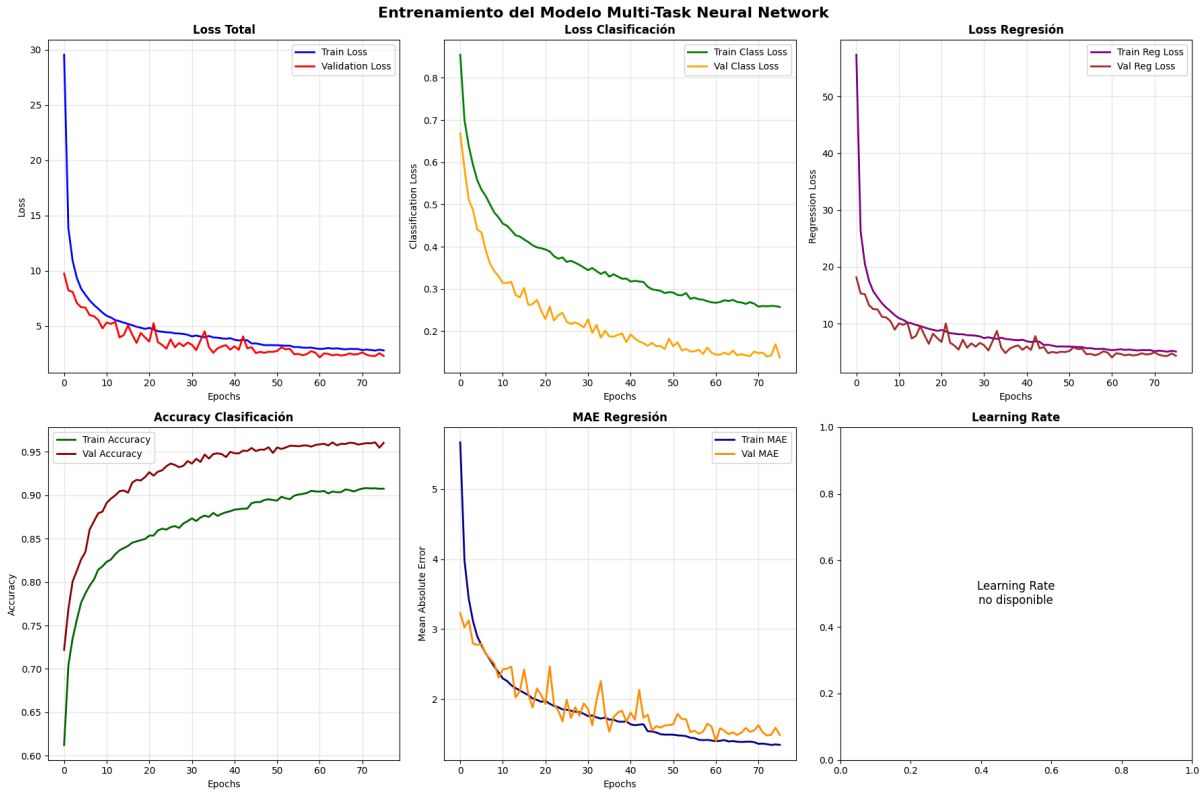
Regression Performance (MMSE Prediction):

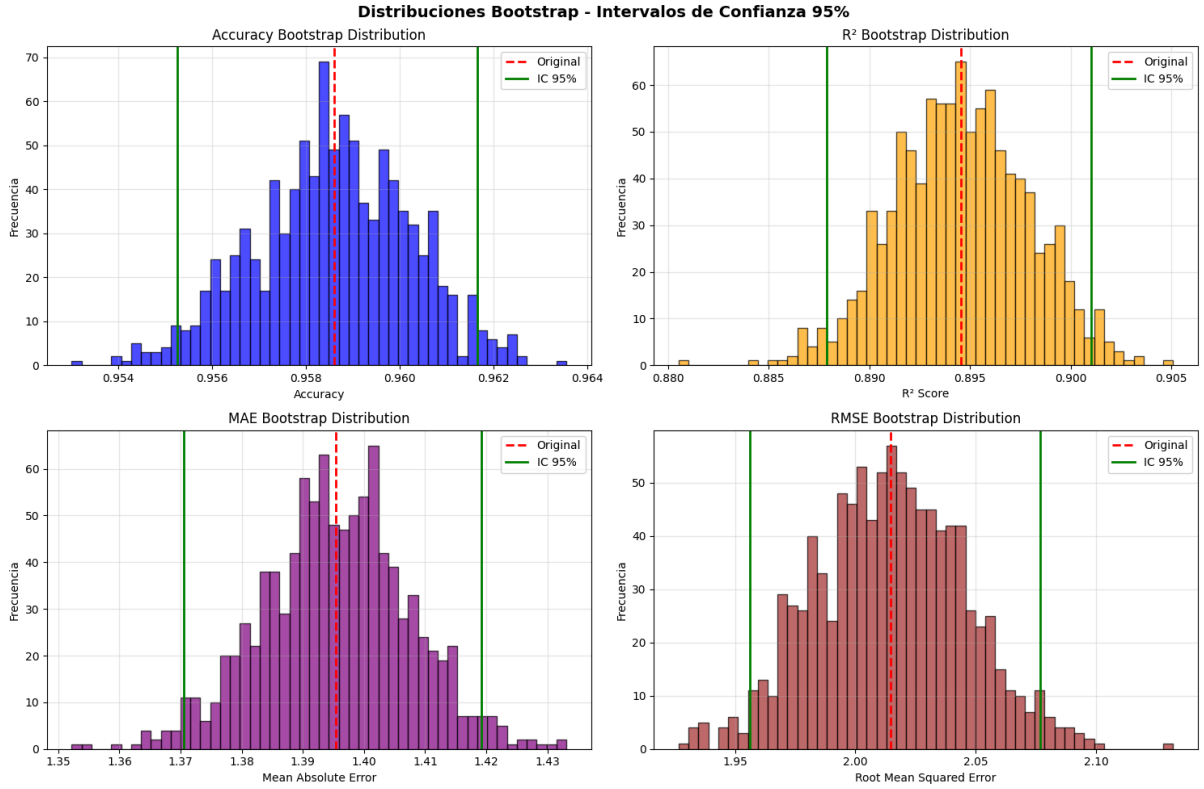
- The model achieved an R^2 score of 0.895 on the test set, indicating that nearly 90% of the variance in MMSE scores is explained by the model.
- The Mean Absolute Error (MAE) was 1.40, and the Root Mean Squared Error (RMSE) was 2.01.
- Bootstrap distributions confirm the robustness of these metrics, with narrow confidence intervals.

Where It Might Fail:

- The model may underperform in edge cases where EEG signals are noisy or atypical (e.g., patients with comorbidities).
- Classification might be less accurate between subtypes of dementia not

present in the dataset or in very early-stage cases with subtle EEG abnormalities.

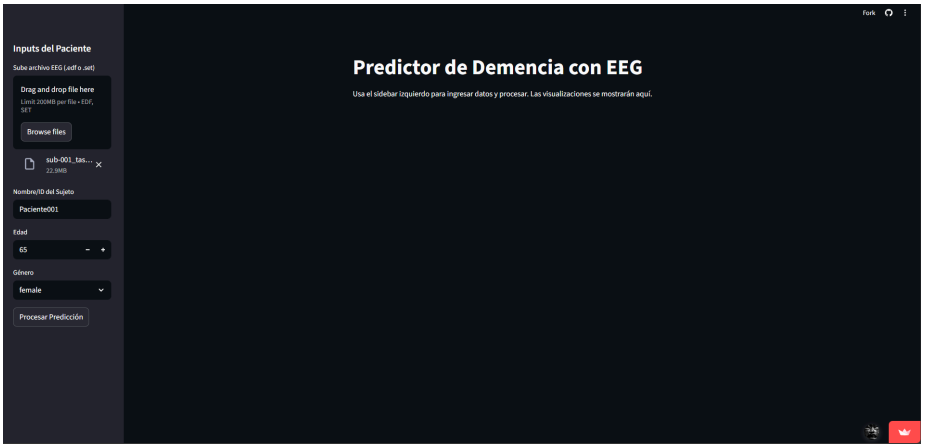




Software

We developed a user-friendly web application where clinicians can upload EEG recordings, input patient age and sex, and obtain immediate predictions of cognitive impairment and disease classification. The app visualizes results and allows report download for clinical use.

Final application



Process

Overview

We spent a significant amount of time handling and preparing the data. The original dataset was not provided in a directly readable format, which led to several challenges during the downloading and parsing stages. Once accessed, we performed data cleaning, formatting, and validation to ensure consistency across all samples.

We also invested time designing and tuning the multi-task model architecture to handle both classification and regression outputs. Initial experiments with simpler models or separated tasks yielded lower performance. Several iterations were required to find the right balance between shared and task-specific layers.

Challenges

Dataset preprocessing was the main challenge because we were unfamiliar with the data. Although it followed the BIDS convention, it required navigating complex folder structures and interpreting metadata formats we hadn't seen before. We had to learn how to properly extract and structure the EEG recordings, which involved understanding domain-specific data organization and working with new Python libraries tailored to EEG and BIDS.

Member Contribution

We worked together throughout the entire project, collaborating on all major components including data preparation, modeling, and evaluation. Andrés took the lead on implementing the neural network architecture and developing the application interface, while both team members contributed jointly to design decisions, testing, and documentation. All key parts of the project were discussed and refined together to ensure consistency and shared understanding.

Conclusion

Limitations

The model's performance is limited by dataset size and diversity, and the lack of external validation on independent datasets. Also, real-world EEG data may have noise or artifacts not present in this dataset, affecting predictions.

Future Improvements

Collecting and training on larger, more diverse datasets would improve generalization. Incorporating advanced artifact removal or adaptive preprocessing pipelines could enhance robustness. Exploring explainable AI techniques could make predictions more transparent to clinicians.

Reflection

This project demonstrated the potential of AI to support neurodegenerative disease diagnosis using EEG. With more data and clinical validation, the tool could become an accessible aid in early detection and monitoring. In hindsight, allocating more time to data augmentation and external validation would strengthen the model's reliability.

Cite

Andreas Miltiadous and Katerina D. Tzimourta and Theodora Afrantou and Panagiotis Ioannidis and Nikolaos Grigoriadis and Dimitrios G. Tsalikakis and Pantelis Angelidis and Markos G. Tsipouras and Evripidis Glavas and Nikolaos Giannakeas and Alexandros T. Tzallas (2024). A dataset of EEG recordings from: Alzheimer's disease, Frontotemporal dementia and Healthy subjects. OpenNeuro. [Dataset] doi: doi:10.18112/openneuro.ds004504.v1.0.8